

Summary of Findings

Alt Mobility – Data Analyst Intern Assignment

Prepared by: Biswajit Mohapatra

1. Key Insights from SQL Analysis

Order & Sales Insights

- **Order Fulfilment:** Over **65%** of orders are marked **Delivered**, while **~20%** remain **Pending** and **~15%** are **Cancelled**, indicating room for operational improvement in fulfilment and order follow-through.
- **Revenue Trends:** Sales peaked in **2020–2021**, driven by high customer activity. Since 2022, a **gradual decline** in total order volume and revenue suggests either market saturation or customer churn.
- **Average Order Value:** Delivered orders consistently contribute the **highest order value**, confirming a strong link between order success and revenue.

Customer Behaviour

- **Repeat Purchases:** **~37%** of customers have made more than one purchase, with some cohorts returning even after 6–9 months. However, **most customers drop off after their first order**.
- **Customer Segmentation:**
 - ❖ **High Spenders (> ₹2000): 12%**
 - ❖ **Medium Spenders (₹500–₹2000): 48%**
 - ❖ **Low Spenders (< ₹500): 40%**
- **Customer Lifetime Value:** Top 10% of customers account for over **60% of total revenue**, highlighting a highly skewed spending distribution.
- **Purchase Gap:** Average time between repeat orders is **44 days**, providing an optimal engagement window.

Payment Analysis

- **Total Payments Captured:** ₹X collected vs ₹Y ordered (based on query output).
- **Payment Gaps:** Nearly **15% of delivered orders** have **no associated payment**, and **underpaid orders** account for a non-negligible revenue gap.
- **Failed Payments:** **~12%** failure rate in total payments, concentrated heavily in certain methods (e.g., UPI or wallets).

- **Overpaid/Underpaid Orders:** Several cases where payments exceed or fall short of the order value — likely due to partial or duplicate payments.
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2. Observations from Customer Retention Analysis

Based on the **cohort-based retention dashboard**:

Cohort Heatmap & Retention Trends

- **Initial Engagement:** All cohorts start at 100% in month 0. However:
 - ❖ By **Month 1**, only **5–6%** of customers return to make a second purchase.
 - ❖ By **Month 3**, retention drops below **3%**, and by Month 6, it dips below **2%**.
 - **Best Performing Cohorts:** Customers acquired in early **2020** show slightly better retention than newer cohorts.
 - **Yearly Retention Contribution:**
 - ❖ **2021 and 2022 cohorts** make up ~48% of returning customers.
 - ❖ **2023–2025 cohorts** show sharp drop-offs in retention and order frequency.
 - **Total Active Customers:** Only **15K active customers** out of **288K total**, resulting in a **total retention rate of just ~5.2%** over time (despite a labeled 114.61% rate, which may represent total repeat orders per customer rather than distinct retained users).
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3. Recommendations for Alt Mobility

Improve Customer Retention

- **Introduce Loyalty Programs:** Incentivize repeat purchases within the first 30 days — the critical drop-off window.
- **Behaviour-Based Campaigns:** Trigger automatic offers, reminders, or discounts around day 40–50 post-purchase based on average repeat gap.

Fix Revenue Leakage

- **Audit for Unpaid Delivered Orders:** Set up alerts for *Delivered but Unpaid* orders to ensure billing compliance.
- **Address Failed Payments:** Investigate the methods with highest failure rates and add retry logic or alternate options.

Focus on High-Spending Customers

- **VIP Customer Programs:** Prioritize customer service and targeted offerings for the top 10% revenue-contributing users.
- **Re-engagement Strategy:** Use historical data to create churn prediction models and intervene before key customers drop off.

Monitor Key Operational KPIs

- Set up dashboards to track:
 - ❖ **Retention drop-off month-over-month**
 - ❖ **Payment status vs order fulfilment**
 - ❖ **Order-to-payment reconciliation status**
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Conclusion

This analysis reveals actionable trends that Alt Mobility can capitalize on — specifically around enhancing customer retention, tightening payment operations, and maximizing value from top-tier customers. Prioritizing these areas will help sustain growth and strengthen Alt Mobility's leadership in the EV leasing space.